

Module 7: Iterators Deep Dive (Advanced X Functions)

7.1 Concept

Iterators like `SUMX` , `AVERAGEX` , `MAXX` , `MINX` , and `RANKX` allow **row-by-row calculations** that go beyond basic aggregation.

- Useful for **ranking, filtering, and conditional aggregations**
- Can be combined with `CALCULATE` for **context-aware calculations**
- Essential for creating **advanced KPIs** in Power BI

7.2 Key Functions Overview

Function	What it Does	Example (on dataset)
<code>SUMX(table, expression)</code>	Row-by-row sum	<code>SUMX(Sales, Sales[Quantity]*Sales[UnitPrice]*(1-Sales[Discount]))</code>
<code>AVERAGEX(table, expression)</code>	Row-by-row average	<code>AVERAGEX(Sales, Sales[Profit])</code>
<code>MAXX(table, expression)</code>	Maximum value row-by-row	<code>MAXX(Sales, Sales[Profit])</code>
<code>MINX(table, expression)</code>	Minimum value row-by-row	<code>MINX(Sales, Sales[Profit])</code>

Function	What it Does	Example (on dataset)
RANKX(table, expression, , order)	Ranks rows based on expression	RANKX(ALL(Sales[Product]), SUM(Sales[SalesAmount]), , DESC)
FILTER(table, condition)	Returns a filtered table	FILTER(Sales, Sales[Profit] > 1000)

7.3 Practical Examples

Example 1: Rank Products by Sales

```
Product Sales Rank :=
RANKX(
    ALL(Sales[Product]),
    SUM(Sales[SalesAmount]),
    ,
    DESC
)
```

- Highest-selling product → Rank 1

Example 2: Top 5 Profitable Products

Top 5 Profitable Products :=

```
CALCULATE(  
    SUM(Sales[Profit]),  
    FILTER(  
        Sales,  
        RANKX(ALL(Sales[Product]), SUM(Sales[Profit]), , DESC) <= 5  
    )  
)
```

- Sums profit for top 5 products only.

Example 3: Category Average Profit (Row-Level)

Avg Profit per Transaction :=

```
AVERAGEX(  
    FILTER(Sales, Sales[Category] = "Electronics"),  
    Sales[Profit]  
)
```

- Returns average profit **only for Electronics** transactions.

Example 4: Max Profit per Category

```
Max Profit by Category :=  
MAXX(  
    FILTER(Sales, Sales[Category] = "Fashion"),  
    Sales[Profit]  
)
```

- Finds the **single highest profit** in Fashion.

7.4 Exercises (Hands-On)

1. Rank Products by Profit

- Measure: Rank all products based on total Profit (descending).

2. Top 3 Products by Sales

- Measure: Sum SalesAmount for **top 3 ranked products**.

3. Average Profit per Electronics Sale

- Measure: Use AVERAGEX with a FILTER for Category = Electronics .

4. Maximum Profit per Fashion Sale

- Measure: Use MAXX to get **highest single profit** in Fashion.

5. Filter High Value Orders

- Measure: Sum SalesAmount for all orders with SalesAmount > 5000 using FILTER and SUMX .

7.5 Visualizations in Power BI

1. Table/Matrix

- Product | Category | SalesAmount | Profit | Product Sales Rank

2. Bar Chart

- X-axis = Product
- Y-axis = SalesAmount for **Top 3 Products**

3. Card Visuals

- Max Profit in Fashion
- Average Profit per Electronics Sale

7.6 Exercise Answers (For Trainers)

1. Rank Products by Profit

Product Profit Rank :=

```
RANKX(  
    ALL(Sales[Product]),  
    SUM(Sales[Profit]),  
    ,  
    DESC  
)
```

- Electronics products like Laptops/Phones → Rank 1–3

2. Top 3 Products by Sales

```
Top 3 Products Sales :=  
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    FILTER(  
        Sales,  
        RANKX(ALL(Sales[Product]), SUM(Sales[SalesAmount]), , DESC) <= 3  
    )  
)
```

- Sum includes top 3 high-selling products

3. Average Profit per Electronics Sale

```
Avg Profit Electronics :=  
AVERAGEX(  
    FILTER(Sales, Sales[Category] = "Electronics"),  
    Sales[Profit]  
)
```

- Typically higher than Fashion/Accessories

4. Maximum Profit per Fashion Sale

```
Max Profit Fashion :=  
MAXX(  
    FILTER(Sales, Sales[Category] = "Fashion"),  
    Sales[Profit]  
)
```

- Returns the **highest profit** recorded in Fashion

5. Filter High Value Orders

```
High Value Sales :=  
SUMX(  
    FILTER(Sales, Sales[SalesAmount] > 5000),  
    Sales[SalesAmount]  
)
```

- Sums all sales >5000, usually large Electronics orders

✓ By the end of Module 7, students will:

- Use iterators (SUMX , AVERAGEX , MAXX , MINX) with **filters**.
- Rank products and sum/filter top performers.
- Calculate **category-specific KPIs** using row-by-row logic.
- Combine iterators with CALCULATE for advanced DAX measures.

This completes the **Advanced DAX Class** with **all 7 modules**, exercises, and answers, fully tied to the **AdvancedDAX_SalesData.csv** dataset.